DDD-P-46D
July 14, 1975
SUPERSEDING
Int. Fed. Spec. DDD-P-0046c(VAM1)
October 14, 1968 and
Fed. Spec. DDD-P-46b
May 7, 1968

FEDERAL SPECIFICATION

PACK, GAUZE, ABDOMINAL

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers an abdominal gauze pack, having an X-ray opaque tab, suitable for use in surgery.

1.2 Classification. The packs covered by this specification shall be of the following sizes, as specified (see 6.2):

Size	Inches (Nominal)
1	4 by 18
2	8 by 36
3	12 by 12
4	18 by 18

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

Federal Specifications:

V-T-276	Thread, Cotton
CCC-G-101	Gauze, Absorbent
DDD-T-86	Tape, Textile; Cotton, General Purpose, (Unbleached, Bleached or Dyed)
MMM-A-250	Adhesive, Water-resistant, (For Sealing Fiberboard Boxes)
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-T-76	Tape, Pressure-sensitive Adhesive Paper, (For Carton Sealing)

FSC 6510

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Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Std. No. 751 - Stitch, Seam and Stitching.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from the Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO; Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Gauze. Gauze shall be bleached white conforming to Type III of CCC-G-101 (see 4.5.1).

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3.1.2 Thread. Thread shall be white cotton No. 60 or 70, 3 cord, type IA2, conforming to V-T-276 (see 4.5.2).

3.1.3 Tape. Tape shall be 3/8-inch minimum width, bleached, white conforming to DDD-T-86 (see 4.5.3). Any of the type weaves specified is acceptable.

3.1.4 X-ray tab. The X-ray tab shall be fabricated of a plastic material and impregnated with an inert nontoxic substance which is opaque to X-rays.

3.2 Design and construction.

3.2.1 Pack. The pack shall be made from four plies of gauze securely sewn together, with an X-ray tab and the ends of a tape loop securely sewn in one corner of the pack. There shall be no exposed raw edges and no open pockets. The pack shall be divided into a number of rectangular sections, "U" sections or squares by means of suitable transverse seams, or transverse and longitudinal seams, as shown in Figure 1. The minimum number of transverse and longitudinal seams, not including the seams at the edge of the pack, shall be as specified in Table I. Where edges are folded-in to prevent exposed raw edges, the edges shall be folded - in approximately 1/2 inch and then sewn. Edge stitching shall be located not less than 1/8 inch from the edge of the pack. Two edges of material may be joined with a row of stitching at least 1/4 inch from the raw edges, and then the joined layers turned inside out to form an inside seam. The X-ray tab and tape loop shall be fastened to the pack by one of the following methods:

a. The tab shall be placed in a corner of the pack, between the layers of gauze (two layers on each side), and secured in a pocket formed by double folding the hem the length of the tab. The tape ends shall be placed in the same corner of the pack and superimposed on the pocket, a minimum length of one inch, in such a manner that they are caught in the double stitching of the tab. The tab shall be secured to the pack by a double row of stitching through the tab for the full length of the tab.

b. In a corner of the pack the tab shall be placed between the layers of gauze (two layers on each side), and secured within the pocket formed by box-x-stitching. The tape ends shall be placed in the same corner of the pack between the layers of gauze, the same as the tab, in such a manner that they will be caught in a minimum of four rows of stitching formed by the box-x-stitch operation.

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c. The tab and the tape ends shall be placed in a corner of the pack between the layers of gauze (two layers on each side). The long dimensions of the tape and the tab shall be parallel, and the tape ends shall be placed on the top of the tab. The tape ends shall be secured with the same stitching used in sewing the edges of the pack. The tab and tape ends shall be further secured to the pack by bartacking along the long dimension for at least 1/2 inch. The bartack shall begin approximately 1/4 inch from end of the tape. For size 4 pack the bartack shall be centrally located on the X-ray tab.

d. The tab shall be placed in a corner of the pack, in a pocket formed by folding over a hem of the short side of the incompletely sewn pack. The tab shall be completely enclosed. The tape ends shall be enclosed in the pocket, to a minimum length of 1 inch. As an alternate, the tape ends shall be placed outside the pocket holding the tab, but still between layers of gauze when the pad is completely sewn. Holding stitching shall be at least 2 lines of stitching through the full length of the tab, holding both tab and tape.

Table I. Number of seams per pack

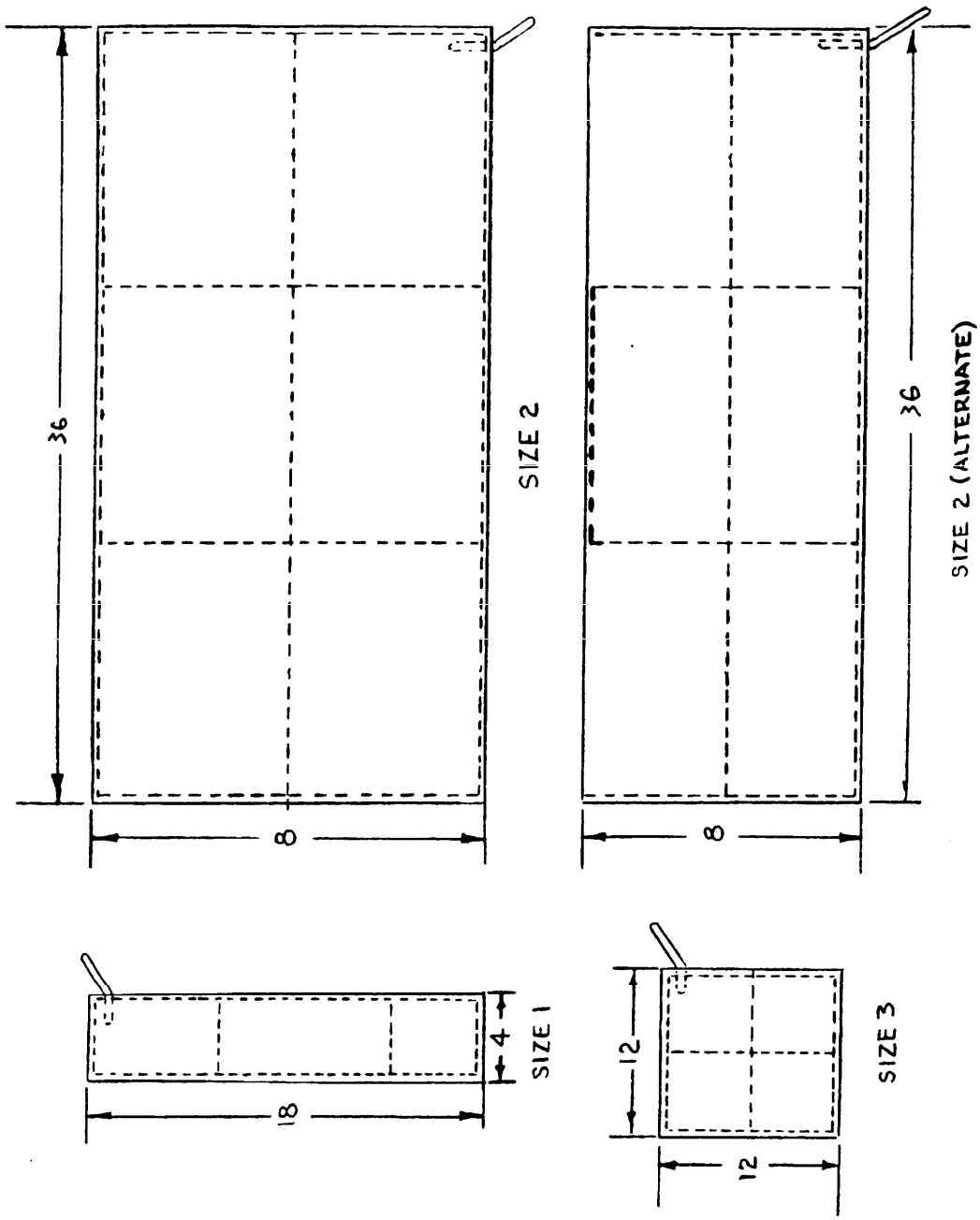
Size	Transverse seams, minimum	Longitudinal seams, minimum
1	0	0
2	2	1
3	1	1
4	1	1

3.2.2 Stitching. Stitching shall be type 301 in accordance with Federal Standard No. 751. The minimum number of stitches per inch shall be six. The thread breaks and ends of seams and stitches shall be back-stitched or backtacked not less than 1/2 inch. Thread breaks in lines of stitching are permitted when they cross a border line of stitching or when the end of the line of stitching meets a border line of stitching at an angle. There shall be not more than two such stitch line breaks per pack.

3.3 Performance.

3.3.1 X-ray opacity. When the pack is covered by 5/8 inch thick aluminum, and subjected to X-ray radiation as specified in 4.5.4, the radiographic image of the X-ray tab on the exposed X-ray film shall be readily visible when viewed by light transmitted through the radiograph.

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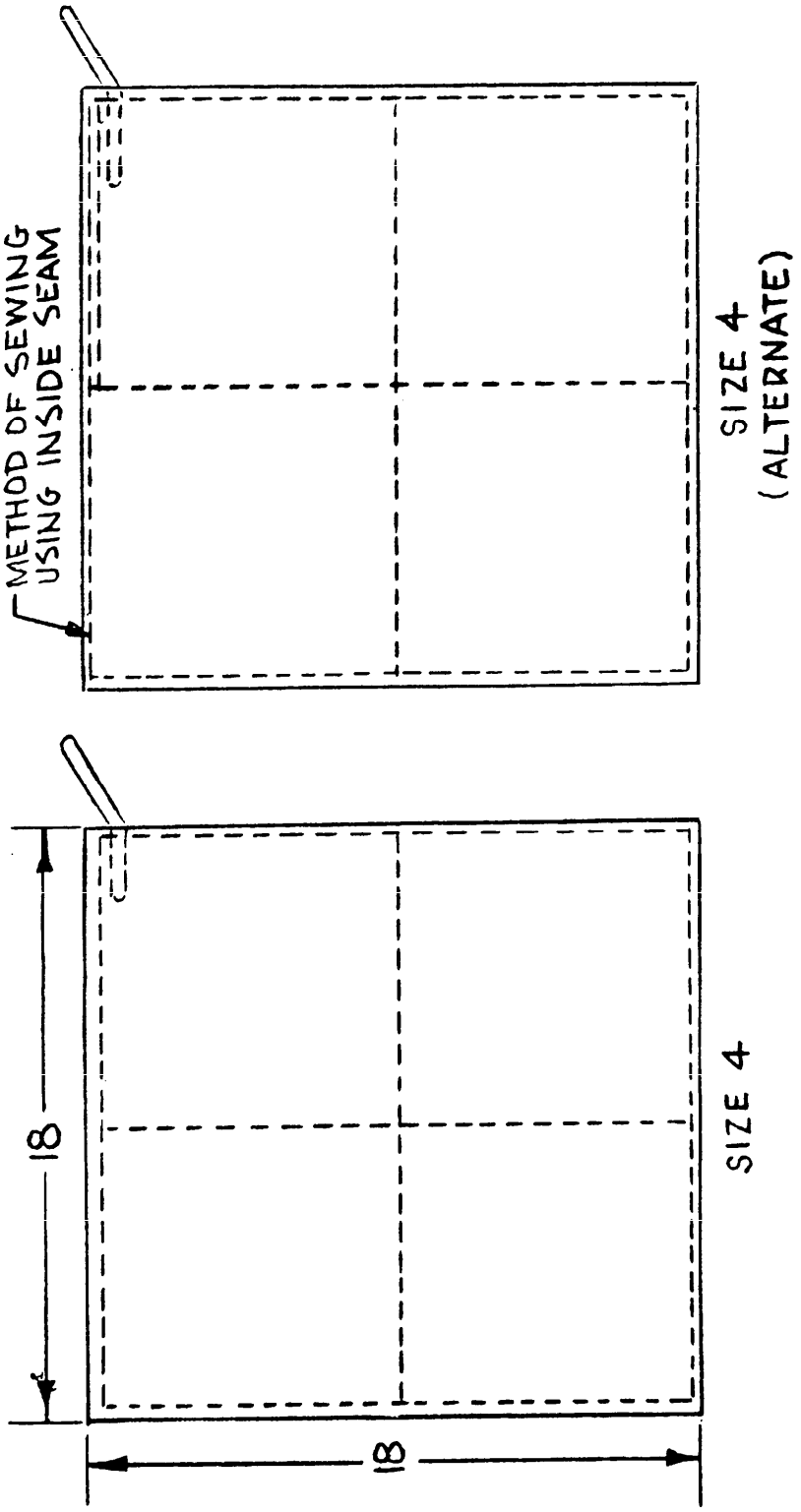


NOTE:
ALL DIMENSIONS IN INCHES

ARRANGEMENT OF SEAMS

FIGURE 1

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ARRANGEMENT OF SEAMS ((ONT'D)

FIGURE 1

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3.3.2 Sterilization. When tested as specified in 4.5.5, the X-ray tab shall not bleed, become tacky or show any other signs of deterioration.

3.4 Dimensions.

3.4.1 Pack. The dimensions of the finished pack shall be in accordance with Table II.

3.4.2 X-ray tab. The tab for sizes 1, 2 and 3 shall measure not less than 1-1/4 inches long, and not less than 5/16 inches wide. The tab for size 4 shall measure not less than 3-1/2 inches long, and not less than 5/16 inches wide.

3.4.3 Tape loop. The tape loop shall extend at least 7 inches from the pack.

3.5 Workmanship. Workmanship shall be first class throughout. The abdominal packs shall be free from all defects that detract from their appearance or may impair their serviceability.

Table II. Dimensions of finished pack

Size	Minimum dimensions for individual pack (inches)			Average minimum dimensions for 1 dozen or more packs (inches)		
1	3-1/2	x	17	3-3/4	x	17-1/2
2	7	x	34	7-1/2	x	35
3	11	x	10	11-1/2	x	11
4	17	x	16	17-1/2	x	17

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Certificates of quality. Certificates of quality supplied by the manufacturer of the materials may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number (NSN), the item identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, the individual test results, and the specification requirements.

4.2 Inspection. Inspection of the abdominal gauze pack shall consist of:

(a) Examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) (see 4.4).
(b) Tests (determination by technical means of physical and chemical properties) (see 4.5).

4.3 Sampling.

4.3.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in Table III. The unit of product for examination purposes shall be a single pack. Not more than 10 packs shall be taken from any single package.

Table III. Sampling for examination

Type defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	4.0
Dimensional	S-3	2.5

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4.3.2 For tests.

4.3.2.1 Raw material. Sampling for raw material tests shall be in accordance with the applicable test specification for each material.

4.3.2.2 X-ray opacity. The unit of product for the X-ray opacity test shall be a single pack. Sample size shall be as indicated for inspection level S-2 in MIL-STD-105, except that the acceptance number shall be zero for all sample sizes, and a minimum of three samples shall be taken.

4.3.2.3 Sterilization. Sample size for the sterilization test shall be three tabs for each lot of radiopaque material. Acceptance number shall be zero.

4.4 Examination.

4.4.1 Dimensional. Examination shall be made to determine compliance with the applicable dimensional requirements. Any dimension that is not within the specified tolerances for dimension, shall be classified as a defect.

4.4.2 For defects. The finished pack shall be examined for defects in material, workmanship, cleanness, and construction as specified in Table IV. The defects shall not be restricted to the defects listed.

Table IV. Classification of defects

Classification	Defect
Major	
101	Hole, cut, tear or burn.
102	Lint.
103	Hanging thread.
104	Loose piece of thread.
105	Thread break, except when permitted, not backstitched or backtacked.
106	Raw edge exposed.
107	Open seam over 1/4 inch.
108	The minimum number of seams per pack not as specified.
109	Tape (tape loop) not secured as specified.
110	X-Ray opaque tab not constructed as specified.
111	X-Ray opaque tab not secured to pack as specified.

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Table IV. Classification of defects (Cont'd)

Classification	Defect
Minor	
201	Spot or stain.
202	Thread end not trimmed.
203	Seam excessively twisted, excessively puckered or pleated.
204	Thread break, backstitched or backtacked less than 1/2 inch.
205	Open seam up to 1/4 inch inclusive.
206	Stitches per inch less than minimum specified.
207	Stitch tension loose, resulting in a loose seam.
208	Stitch tension tight, causing excessive puckering.
209	Overlap greater than 3/8 inch, when two edges of (folded) gauze are sewn together.
210	Tape loop not formed as specified (including minimum length).

4.4.3 Preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of section 5. Defects shall be scored as specified in Table V. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

Table V. Examination of preparation for delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and packing materials	Any component missing or damaged
Workmanship	Inadequate application of components such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

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4.5 Tests.

4.5.1 Gauze. Gauze shall be tested in accordance with CCC-G-101 (see 3.1.1).

4.5.2 Thread. Thread shall be tested in accordance with V-T-276 (see 3.1.2).

4.5.3 Tape. Tape shall be tested in accordance with DDD-T-86 (see 3.1.3).

4.5.4 X-ray opacity. An aluminum step-wedge ranging in thickness from 1/8 to 3/4 inch in steps of 1/8 inch, shall be placed on an unexposed X-ray film. X-ray exposure shall be approximately 10 MAS, 80 KVP at 30 inches using medium speed or detail screens in a convenient cassette. The X-ray exposure shall be such that the exposed, developed film showing the image of the step wedge shall indicate a clear delineation of at least the 1/4 inch, 3/8 inch, 1/2 inch and 5/8 inch steps. The corner of the pack (single layer) containing the X-ray tab shall be placed on an exposed X-ray film and covered by 5/8 inch thick aluminum. When exposed to X-rays of the same intensity as specified above, it shall meet the requirements of 3.3.1.

4.5.5 Sterilization. The tab shall be kept in a sterilizer at a temperature of 250° F., under live steam conditions, for one hour and shall meet the requirements of 3.3.2.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.2).

5.1.1 Unit of issue. One package containing one hundred abdominal gauze packs of the size specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package. One hundred abdominal packs of one size only shall be neatly packaged in a bag fabricated from virgin polyethylene seamless tubing and having an average wall thickness of not less than 0.004 inch with no wall thickness at any point below 0.0033 inch. Excess air shall be expelled or evacuated from the bag and closure effected by a continuous heat seal.

5.1.2.2 Intermediate package. Three unit packages of one size only shall be packaged in a snug fitting box constructed in accordance with PPP-B-636, Class weather-resistant. Box shall be closed by taping the top and bottom flaps with 2 inch wide tape conforming to PPP-T-76. Closing of bottom flaps with adhesive conforming to MMM-A-250 or metal stitches is permitted.

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5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B, or C as specified (see 6.2).

5.2.1 Level A. Two intermediate packages (6 units) of Size 2 or 4, four intermediate packages (12 units) of Size 3 or six intermediate packages (18 units) of size 1 abdominal packs shall be packed in an exterior container designed for a Type 2 load and constructed in accordance with PPP-B-636, Class weather-resistant, Grade V3c or V3s. Closure and strapping shall be as specified in the appendix of PPP-B-636.

5.2.2 Level B. Abdominal packs shall be packed as specified in paragraph 5.2.1 except that the exterior container shall be constructed in accordance with PPP-B-636, Class domestic, Grade 275. Closure of the box shall conform to method II as specified in the appendix of PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, constructed so as to insure acceptance and safe delivery by common or other carriers to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil Agencies. In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with Federal Standard No. 123.

5.3.2 Military Activities.

5.3.2.1 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. In addition, marking shall include name and address of manufacturer, lot (control) number, date of manufacture in lieu of date packed, and the requirements of the Federal Food, Drug, and Cosmetic Act. The required marking shall be printed either directly on the bag or on a label inserted within the bag in such a manner as to permit ready identification.

5.3.2.2 Intermediate package. Each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, water-proofing shall be required only when applicable carton is fabricated of water-resistant material. Lot (control) number, contract or purchase order number, and name of contractor shall be shown. Date of manufacture shall be shown in lieu of date packed.

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5.3.2.3 Exterior containers. Exterior containers shall be marked in accordance with MIL-STD-129. Lot (control) number shall be shown. Date of manufacture shall be shown in lieu of date packed.

6. NOTES

6.1 Intended use. The abdominal gauze pack covered by this specification is intended to be used in surgery.

6.2 Ordering data. Procurement documents should specify:

(a) Title, number, and date of this specification.
(b) Size of packs required (see 1.2).
(c) National Stock Number.
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification is intended to cover the following items appearing in the Federal Supply Catalog:

National Stock No.	Item Identification	Size
6510-00-202-8000	PACK, GAUZE, ABDOMINAL, Radiopaque, 4 by 18 inches, 100s	1
6510-00-202-8530	PACK, GAUZE, ABDOMINAL, Radiopaque, 8 by 36 inches, 100s	2
6510-00-202-8555	PACK, GAUZE, ABDOMINAL, Radiopaque, 12 by 12 inches, 100s	3
6510-00-559-3173	PACK, GAUZE, ABDOMINAL, Radiopaque, 18 by 18 inches, 100s	4

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Military Custodians:

Army - MD
Navy - MS
Air Force - 03

Civil Agency Coordinating Activities:

GSA-FSS
VA-DMS
HEW-FDA

Preparing activity:

DSA-DM

Project No. 6510-0609

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DOCUMENT IDENTIFIER AND TITLE
DDD-P-46D PACK, GAUZE, ABDOMINAL

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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